CLASS 9 — MATHS · FORMULA SHEET

Chapter 7: **Probability** | Quick revision, one page

Probability

- Theoretical probability: $P(E) = \dfrac{\text{number of outcomes favourable to } E}{\text{total number of equally likely outcomes}}$.
- Experimental probability: $P(E) = \dfrac{\text{number of times } E \text{ occurred}}{\text{total number of trials}}$.
- Always $0 \le P(E) \le 1$.

Special events

- Sure event: $P = 1$. Impossible event: $P = 0$.
- Complement: $P(\text{not } E) = 1 - P(E)$, i.e. $P(E) + P(\text{not } E) = 1$.
- The sum of the probabilities of all outcomes of an experiment is 1.

Sample spaces

- One coin: $\{H, T\}$. Two coins: $\{HH, HT, TH, TT\}$ (4 outcomes). Three coins: 8 outcomes.
- One die: $\{1, 2, 3, 4, 5, 6\}$. Two dice: $6 \times 6 = 36$ outcomes.
- Cards: 52 cards, 4 suits of 13; 26 red and 26 black.

Estimation

- Expected number of times E occurs in N trials $\approx N \times P(E)$.
- More trials make the experimental probability closer to the theoretical probability.

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